

# IT-SIMPLERA INSTITUTE

## Cyber Security Analyst Internship Program



## INTERNSHIP WEEKLY REPORT

Week 01 Cybersecurity Fundamentals & Security Concepts

|                 |                       |
|-----------------|-----------------------|
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| Week            | Week 01               |
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## Executive Summary

During Week 01 of the IT-SIMPLERA Cyber Security Analyst Internship Program. The primary focus of this week was establishing a strong foundation in cybersecurity fundamentals by researching real-world cyberattacks, analyzing their impact through the lens of the CIA Triad model, exploring industry-standard security tools, and designing a secure home network architecture.

During the practical component of this week, a controlled penetration testing lab environment was configured using VirtualBox, featuring Kali Linux as the attacker platform and Metasploitable2 as the intentionally vulnerable target. Key technical activities included network discovery and port scanning using Nmap, real-time packet capture and traffic analysis with Wireshark, and web service enumeration via curl. The practical phase culminated in the successful exploitation of the Samba usermap\_script vulnerability using the Metasploit Framework, successfully yielding a root-level interactive shell. All practical exercises were conducted strictly within an isolated host-only network environment and are fully documented with supporting terminal output and screenshots throughout this report.

## Learning Objectives

- ❖ **Understand and apply the CIA Triad** (Confidentiality, Integrity, Availability) model to classify and analyze real-world cyberattack scenarios.
  - **Research and document five major real-world cyberattacks**, identifying their attack vectors, organizational impacts, and financial consequences.
  - **Gain operational familiarity with core cybersecurity tools**, analyzing the utility of Nmap for reconnaissance, Wireshark for traffic analysis, and Splunk for enterprise log auditing.
  - **Design a secure home network architecture** that incorporates layered security controls, perimeter firewalls, and strict VLAN network segmentation.
  - **Analyze the daily responsibilities of a Cybersecurity Analyst**, mapping out standard enterprise workflows for alert triage, threat hunting, and incident response.
  - **Configure and operate a penetration testing lab environment** using VirtualBox, deploying Kali Linux as the attack platform and Metasploitable2 as the vulnerable target.

## Environment Setup

| Component               | Details                                     |
|-------------------------|---------------------------------------------|
| Operating System (Host) | Windows (Host Machine)                      |
| Virtualization Software | Oracle VirtualBox                           |
| Attacker VM             | Kali Linux                                  |
| Target VM               | Metasploitable2                             |
| Network Type            | Host-Only Adapter (isolated lab network)    |
| Tools Installed         | Kali Linux                                  |
| Network Interface       | eth0 on Metasploitable2; eth0 on Kali Linux |

# TASK 1: FIVE REAL-WORLD

## CYBERATTACKS

The following five cyberattacks represent significant events in the history of information security. Each incident demonstrates how threat actors exploit vulnerabilities across different sectors and technologies.

| Attack Name & Year         | Target Organization               | Attack Type                    | Description                                                                                                                                                  | Operational & Financial Impact                                                                                                                                                                                     |
|----------------------------|-----------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Change Healthcare (2024)   | Change Healthcare (United States) | Ransomware & Data Exfiltration | The BlackCat (ALPHV) ransomware group used stolen credentials to bypass MFA, encrypting administrative systems and exfiltrating over 6 TB of sensitive data. | Completely crippled U.S. healthcare billing for weeks, preventing prescriptions from filling. Parent company UnitedHealth Group ultimately paid a <b>\$22 million</b> ransom and faced billions in recovery costs. |
| Stryker Corporation (2026) | Stryker Corporation (Global)      | Malware / Device Wipe          | The Iranian-linked hacking group <i>Handala</i> breached the medical device manufacturer's                                                                   | Heavily disrupted worldwide operations, freezing order processing, supply chains,                                                                                                                                  |

| Attack Name & Year           | Target Organization       | Attack Type                     | Description                                                                                                                                                            | Operational & Financial Impact                                                                                                                                       |
|------------------------------|---------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                              |                           |                                 | systems, pushing an update that triggered simultaneous factory resets on over <b>200,000 corporate devices</b> across 79 countries.                                    | production lines, and shipping logistics for days.                                                                                                                   |
| <b>MGM Resorts</b><br>(2023) | MGM Resorts International | Social Engineering / Ransomware | The <i>Scattered Spider</i> group used voice-phishing (vishing) to trick an IT helpdesk employee into resetting credentials, granting them entry to deploy ransomware. | Shut down digital hotel keys, slot machines, ATMs, and reservation systems across Las Vegas for days, costing MGM an estimated <b>\$100 million</b> in lost revenue. |
| <b>SolarWinds</b><br>(2020)  | SolarWinds (Supply Chain) | Supply Chain / Trojan           | Russian state-backed hackers injected malicious code into software updates for SolarWinds'                                                                             | Compromised networks across multiple U.S. federal agencies (like Treasury and Homeland Security) and over                                                            |

| Attack Name & Year       | Target Organization               | Attack Type | Description                                                                                                                                                 | Operational & Financial Impact                                                                                                                                          |
|--------------------------|-----------------------------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |                                   |             | Orion platform, which unsuspectingly went out to thousands of global clients.                                                                               | 100 private corporations, exposing highly sensitive government communications.                                                                                          |
| Colonial Pipeline (2021) | Colonial Pipeline (United States) | Ransomware  | The <i>DarkSide</i> group compromised an old, legacy VPN account lacking multi-factor authentication, deploying ransomware onto corporate billing networks. | Forced a voluntary shutdown of a pipeline carrying 45% of the East Coast's fuel supply, causing widespread panic-buying, fuel shortages, and spikes in gasoline prices. |

## Task 2: CIA Triad Analysis

### UNDERSTANDING THE CIA TRIAD

The **CIA Triad** is the definitive of cybersecurity. Every firewall rule, password policy, and encryption algorithm exist to protect at least one of these three core pillars. When an organization suffers a data breach, it is always measured by which parts of the triad were broken

| Core Component  | Primary Goal          | Meaning in Plain English                                                                                                                             | Real-World Failure Example                                                                                                                     | Common Defensive Control                                                                                  |
|-----------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Confidentiality | Keep secrets secret.  | Ensuring that sensitive data is only accessible to authorized individuals and remains completely hidden from unauthorized eyes.                      | A hacker exploits a weak password to steal and leak a database containing millions of customer social security numbers and credit cards.       | Multi-Factor Authentication (MFA), Data Encryption (at rest and in transit), Access Control Lists (ACLs). |
| Integrity       | Keep data accurate.   | Ensuring that data is trustworthy, correct, and has not been altered, tampered with, or deleted mid-transit or on a system by an unauthorized party. | A cybercriminal intercepts an electronic bank wire transfer and alters the routing numbers to divert \$10,000 into their own offshore account. | Cryptographic Hashing (e.g., SHA-256), Digital Signatures, File Integrity Monitoring (FIM).               |
| Availability    | Keep systems running. | Ensuring that data, networks, and critical business applications are up, running, and accessible to                                                  | A competitor launches a Distributed Denial of Service (DDoS) attack, flooding an e-commerce                                                    | Redundant Backups, Load Balancers, DDoS Mitigation                                                        |

|  |  |                                           |                                                                   |                                    |
|--|--|-------------------------------------------|-------------------------------------------------------------------|------------------------------------|
|  |  | authorized users whenever they need them. | website with fake traffic and crashing the store on Black Friday. | Services, Disaster Recovery Plans. |
|--|--|-------------------------------------------|-------------------------------------------------------------------|------------------------------------|

## TASK 3: CYBERSECURITY TOOLS

The following three cybersecurity tools are industry-standard instruments used by security professionals for network reconnaissance, traffic analysis, and penetration testing. Each tool was practically used during this week's lab exercises.

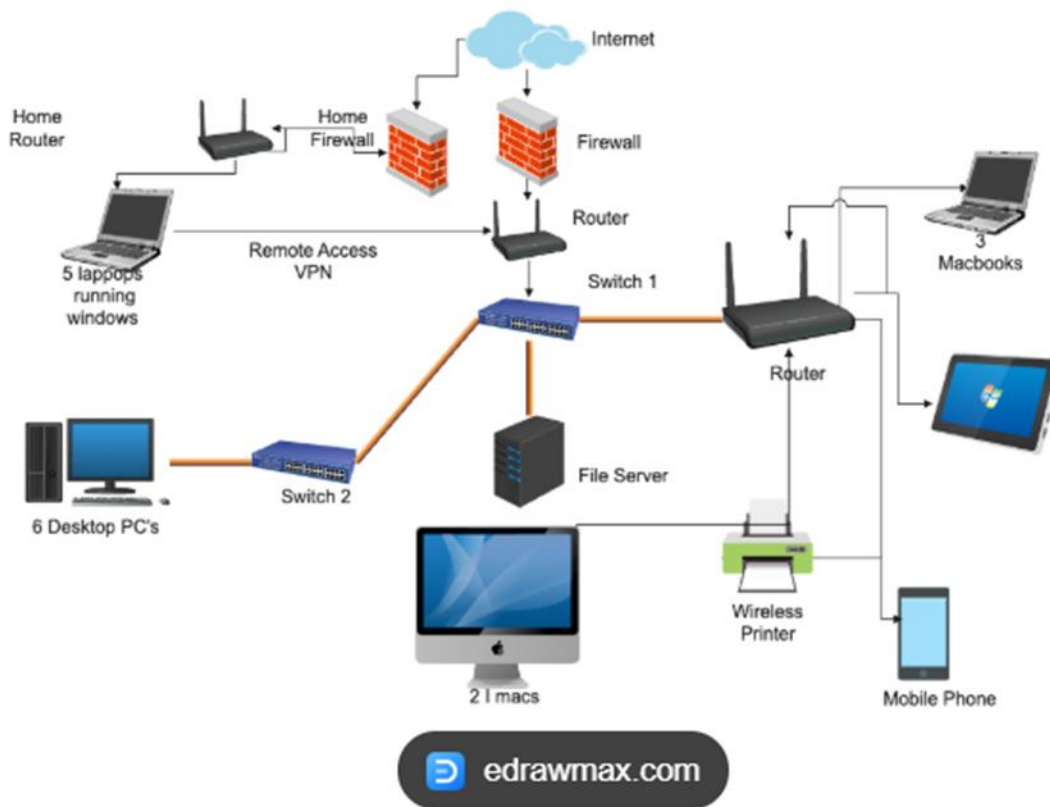
| Cybersecurity Tool    | Purpose                                        | Function                                                                                         | Use Case                                                                                                            |
|-----------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Nmap (Network Mapper) | To discover devices and services on a network. | Scans IP addresses, detects open ports, identifies operating systems, and maps network topology. | Used by network administrators and security professionals to identify vulnerabilities and monitor network security. |
| Wireshark             | To analyze network traffic.                    | Captures and inspects data packets traveling across a network in real time.                      | Used for troubleshooting network issues, detecting suspicious activity, and investigating cyberattacks.             |



|                      |                                                     |                                                                                  |                                                                                                                                  |
|----------------------|-----------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Metasploit Framework | To test system security by simulating cyberattacks. | Identifies, exploits, and validates vulnerabilities in systems and applications. | Used by penetration testers and ethical hackers to assess security and improve defenses before attackers can exploit weaknesses. |
|----------------------|-----------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|

## Task 4: Secure Home Network Design

A secure home network must implement a layered defense-in-depth architecture. The following diagram and explanation describe a comprehensive secure home network design that protects against common threats including unauthorized access, malware, eavesdropping, and data interception.



## Task 5: What Does a Cybersecurity Analyst Do Every Day?

### Introduction

A Cybersecurity Analyst is responsible for protecting an organization's computers, networks, systems, and sensitive data from cyber threats. They monitor security systems, identify risks, respond to incidents, and ensure that the organization's digital assets remain secure.

### Monitoring Security Systems

A Cybersecurity Analyst starts the day by monitoring security alerts from firewalls, antivirus software, intrusion detection systems (IDS), and SIEM platforms. They investigate suspicious activities, unauthorized login attempts, and unusual network traffic to identify potential threats.

## **Vulnerability Assessment**

Analysts regularly scan systems and networks using tools such as Nmap to identify security weaknesses. They recommend software updates, install security patches, and ensure that vulnerabilities are fixed before attackers can exploit them.

## **Incident Response**

When a cyberattack or security incident occurs, the analyst investigates the issue, identifies the source of the attack, isolates affected systems, removes malware, and helps restore normal operations. They also prepare reports and suggest preventive measures.

## **Network and Log Analysis**

Cybersecurity Analysts examine network traffic using tools like Wireshark and review system logs to detect malicious activities. This helps them identify threats early and improve the organization's security posture.

## **User Access Management**

They manage user accounts and access permissions, ensuring that only authorized users can access sensitive information. They also implement security measures such as Multi-Factor Authentication (MFA) and enforce strong password policies.

## **Security Awareness**

Analysts educate employees about cybersecurity best practices, including recognizing phishing emails, creating strong passwords, and following company security policies. Employee awareness helps reduce security risks caused by human error.

## **Continuous Learning**

Cybersecurity is constantly evolving, so analysts regularly learn about new threats, vulnerabilities, and security technologies. They stay updated through training, certifications, and threat intelligence reports.

## **Conclusion**

A Cybersecurity Analyst plays a critical role in protecting an organization from cyber threats. Their daily responsibilities include monitoring systems, responding to incidents, managing vulnerabilities, analyzing network activity, and educating users. Their work helps ensure the confidentiality, integrity, and availability of organizational data and systems.